

Replicated MD identifies persistent nonlocal tag contacts as the most discriminating dynamic readout

Broad 3 × 20 ns screening and independent corrected-protocol validation reproduce the main candidate-control differences

a

Replicated MD design and workload

BROAD SCREEN

13 systems
× 3 independent replicas
× 20 ns
= 39 trajectories | 780 ns



INDEPENDENT CORRECTED VALIDATION

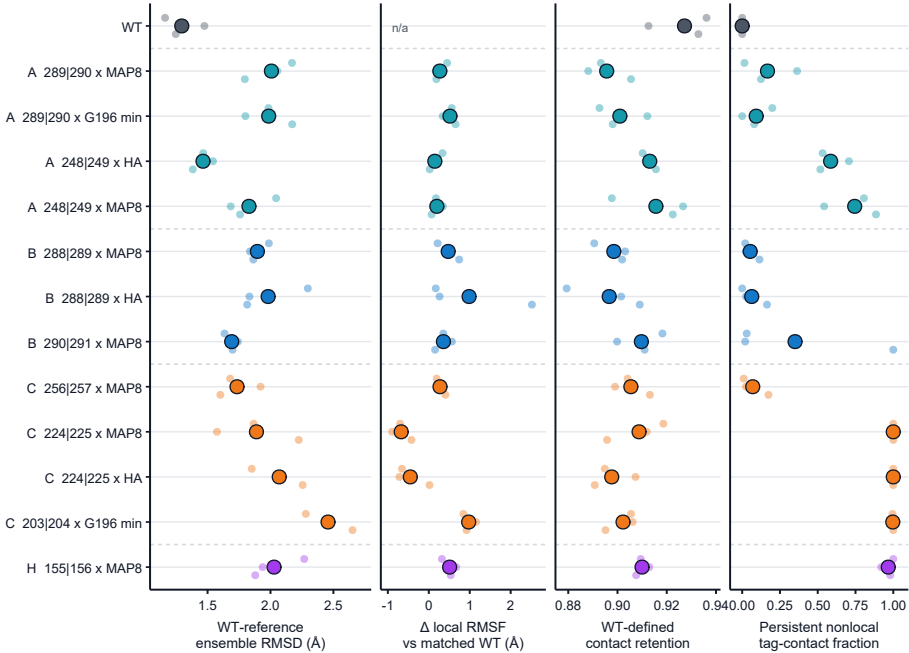
6 systems
× 3 independent replicas
× 20 ns
= 18 trajectories | 360 ns

57 trajectories
1.14 μ s cumulative sampling

screening/validation sampling;
not mechanistic convergence

Broad corrected-MD perturbation landscape

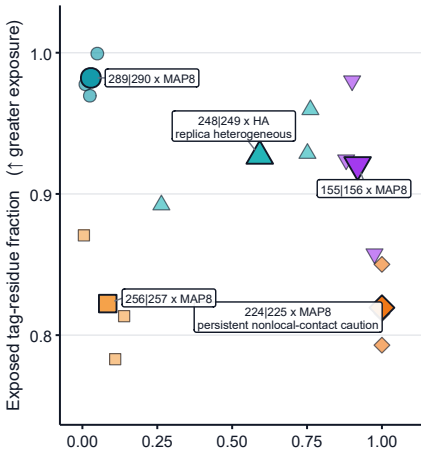
Small translucent dots: replicas; outlined circles: construct means. A/B/C/H denote Priority A, Priority B, conflict control, and hard negative



c

Corrected-protocol replica-level tag behavior

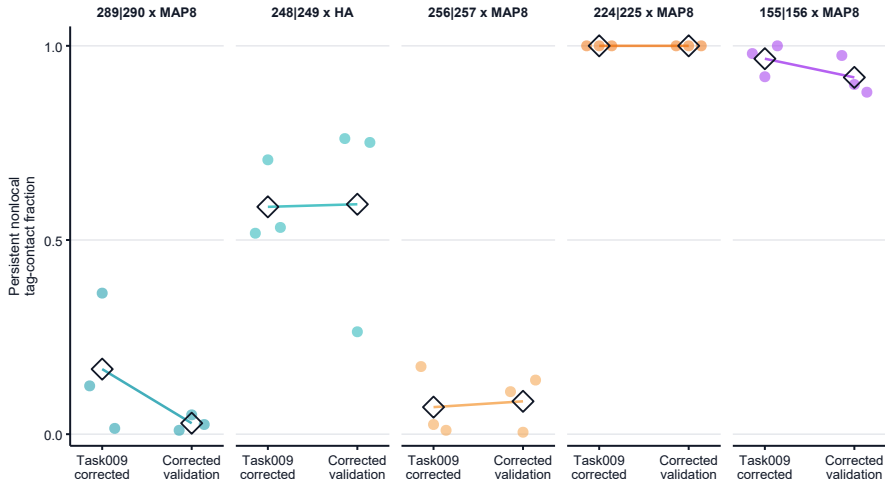
Points: replicas; outlined symbols: means.
No favorable quadrant is defined.



d

Independent corrected protocol reproduces the main contact pattern

Independent 3 × 20 ns ensembles; replicas are not pooled.
Lines connect dataset means for visual direction only.



lower nonlocal contact ← Persistent nonlocal tag-contact fraction